Problem 4

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Problem 1 *For each of the curves given by the following equations, find a formula for the slope of the tangent line at a point (x, y).*

(a) $e^{x^2y^3} - 5x + 7y = 36$

Explanation.

$$e^{x^2y^3}\left(2xy^3 + x^2(3y^2)\frac{dy}{dx}\right) - 5 + 7\frac{dy}{dx} = 0$$

$$2xy^3e^{x^2y^3} + 3x^2y^2e^{x^2y^3}\frac{dy}{dx} - 5 + 7\frac{dy}{dx} = 0$$

$$3x^2y^2e^{x^2y^3}\frac{dy}{dx} + 7\frac{dy}{dx} = -2xy^3e^{x^2y^3} + 5$$

$$\left(3x^2y^2e^{x^2y^3} + 7\right)\frac{dy}{dx} = -2xy^3e^{x^2y^3} + 5$$

$$\frac{dy}{dx} = \frac{-2xy^3e^{x^2y^3} + 5}{3x^2y^2e^{x^2y^3} + 7}$$

(b) $7 = 22\tan(y) + \frac{4}{x} - \frac{7}{y}$

Explanation.

$$0 = 22\sec^2(y)\frac{dy}{dx} - \frac{4}{x^2} + \frac{7}{y^2}\frac{dy}{dx}$$

$$22\sec^2(y)\frac{dy}{dx} + \frac{7}{y^2}\frac{dy}{dx} = \frac{4}{x^2}$$

$$\frac{dy}{dx} = \frac{\frac{4}{x^2}}{22\sec^2(y) + \frac{7}{y^2}}$$

(c) $\cos(xy) - x^3 = 5y^3$

Explanation.

$$-\sin(xy)\cdot\left(y + x\frac{dy}{dx}\right) - 3x^2 = 15y^2\frac{dy}{dx}$$

$$-y\sin(xy) - x\sin(xy)\frac{dy}{dx} - 3x^2 = 15y^2\frac{dy}{dx}$$

$$x\sin(xy)\frac{dy}{dx}+15y^2\frac{dy}{dx}=-y\sin(xy)-3x^2$$
$$\frac{dy}{dx}=\frac{-y\sin(xy)-3x^2}{x\sin(xy)+15y^2}$$

Provided that $x\sin(xy)+15y^2\neq 0$. It is worth pointing out that equivalent condition was not necessary for parts (a) and (b) because the denominator of those solutions cannot be 0.